

Supplementary File of Evaluating Cooperative Resilience in Multiagent Systems: A Comparison Between Humans and LLMs

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December 12, 2025

This supplementary file provides an extended and detailed exploration of some aspects that complement the main paper. While the core content of the paper is fully understandable on its own, this supplementary material offers complementary information into the following items:

- Detailed description of the **Commons Harvest** environment and the game interface used by human participants.
 - Additional information about the human participant groups and recruitment procedure.
 - Explanation of the LLM-based agent architecture.
 - Extended figures and analysis of the collective welfare indicators considered in the study.
 - Description of the construction of the baseline conditions used in the experiments.
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1 Experimental Setup

1.1 Environment description

The study uses the Commons Harvest Open scenario from the Melting Pot 2.0 suite [S1]. In this environment, multiple agents share a confined grid with apple-bearing trees. [Figure S1](#) depicts representative timesteps. At initialization, there are four full trees with 13 apples each and two partial trees in the upper corners with 6 apples each ([Figure S1a](#)). We consider three agents tasked with harvesting apples across the map; for instance, [Figure S1b](#) shows a state after some apples have been collected. Apples regenerate stochastically each step, with a probability that decreases as the tree’s remaining apples drop. If a tree is fully depleted,

^{*}This work was supported by Google through the Google Research Scholar Program, and by the UniAndes–DeepMind Scholarship 2023.

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[‡]This document provides supplementary material accompanying the main manuscript “Evaluating Cooperative Resilience in Multiagent Systems: A Comparison Between Humans and LLMs.”

it disappears permanently. This creates a social dilemma: overharvesting a tree eliminates future yield, harming the group. Thus, agents must regulate harvesting to preserve long-term availability. Agents move in the four cardinal directions and perceive the world through a limited, position-dependent field of view that includes other agents, trees, and walls.

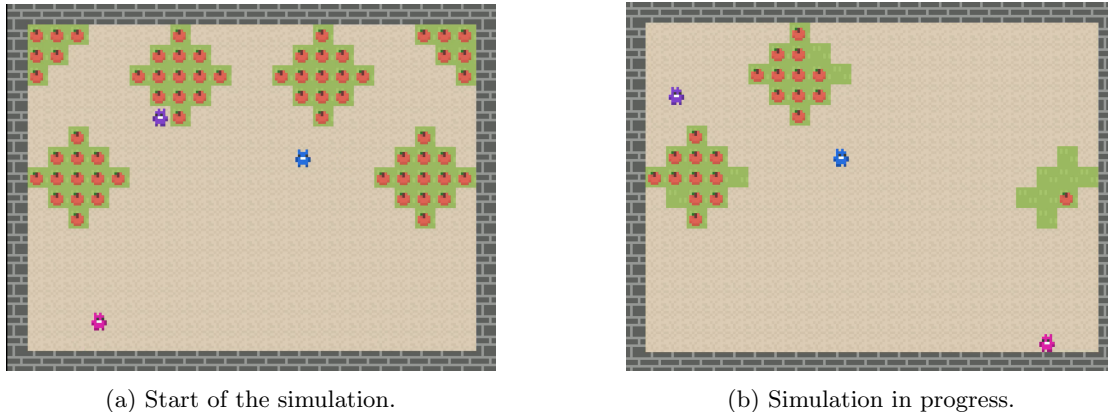


Figure S1: Diagram of the “Commons Harvest Open” environment. In the figure (a) the trees contain all the apples, in (b) the agents have already consumed some apples.

1.2 Human Experiments

The study was approved by the Institutional Review Board and Ethics Committee of the School of Engineering at Universidad de los Andes., Bogotá, Colombia; **Ethical Approval ID 298, issued on July 18, 2025** (see [Section 4](#)). Eligible participants were young adults (≥ 18 years), university students at undergraduate or graduate level, with normal or corrected-to-normal vision; no additional exclusion criteria were applied. Ten individuals completed the nine short-horizon scenarios, and the same ten performed the communication variant. An additional ten participants took part in the extended-horizon human simulations. For the baseline condition, a separate cohort of twenty participants was recruited for both the short and extended settings. In total, forty human subjects participated. Participants were randomly paired in each session.

All participants connected to a remote interface, detailed in [Figure S2](#), from which they controlled their assigned agent and viewed the environment through that agent’s partially observable field of view. They were briefed on the game objective—“Your individual goal is to consume as many apples as possible”—and informed about the apple-regeneration process, as well as the available actions (movement and attack).

For the communication variant, participants used a structured messaging panel with six predefined categories, each mapped to a keyboard key:

- Q** : Information Message
- W** : Question Message
- E** : Individual Strategy
- R** : Collective Strategy
- T** : Establish Agreement
- Y** : Evaluate/Discuss Agreement

Before starting, the experimenter provided examples for each category and verified comprehension; participants proceeded only after confirming they understood the game. All human participants signed an informed consent form acknowledging that the communication sessions would be recorded. They were informed about the ethical guidelines governing the study and explicitly confirmed their agreement to participate under those conditions.



Figure S2: Human-agent interaction interface. Graphical interface used in the human experiments. The Game area (top left) displays the environment from the participant’s assigned agent perspective. The Information area (top right) provides contextual and status updates. The Movement control panel (bottom left) allows navigation and rotation through arrow and rotation keys. The Communication panel (bottom right) contains six predefined message categories—Information (Q), Question (W), Individual Strategy (E), Collective Strategy (R), Establish Agreement (T), and Evaluate/Discuss Agreement (Y)—used during the communication condition.

The system setup and onboarding required approximately 20 minutes. Afterward, baseline groups completed two short runs: 5 minutes without communication and 5 minutes with communication. Groups assigned to the nine-scenario condition spent 45 minutes in total, with communication and no-communication conducted in separate sessions. The extended-horizon variant lasted 2 hours.

1.3 LLM-Based Agents

1.3.1 Architecture CooperativeGPT

The LLM-based agents build on the CooperativeGPT cognitive architecture, an autonomous multi-agent system deployed in a 2D grid environment. Each agent runs a full cognitive cycle: Perceive → Plan → Act → Communicate → Reflect → Execute, driven by a large language model and supported by three complementary memory systems: short-term memory, long-term semantic memory, and spatial memory.

At every timestep, the environment provides each agent with local observations, rewards, and state changes. The Perceive module filters observations according to an attention bandwidth (e.g., closest objects and agents), updates the spatial map, tracks other agents and relevant resources, and writes a narrative “experience” to long-term memory (LTM) with associated metadata (time, type, poignancy). It also invokes an LLM-based react decision that determines whether the agent should replan based on the new information.

If replanning is required, the Plan module queries recent perceptions and reflections from LTM and short-term memory (STM), together with the world rules and the agent biography, to generate a high-level plan and a set of explicit goals (e.g., collect apples in a safe region while avoiding conflict). The Act module then translates this plan into a queue of high-level actions (e.g., move to a position, harvest from a tree, explore unknown regions), which the spatial memory converts into low-level navigation steps in the grid. These steps are executed sequentially by the environment.

The agents maintain three memory subsystems:

- **Short-term memory (STM):** an in-memory store of the current cognitive state (plan, goals, current observations, last actions, known agents, active agreements, etc.) that is passed to all cognitive modules.
- **Long-term memory (LTM):** a vector database (ChromaDB) that stores perceptions, reflections, and communication events as embedded documents, with recency-, poignancy-, and similarity-based retrieval for downstream reasoning.
- **Spatial memory:** a 2D map that tracks explored cells, object locations, other agents, and path-planning information, and that supports conversion from high-level actions to low-level movement sequences.

On top of this core architecture, the system implements a structured communication layer. Agents exchange short (≤ 15 -word) messages through a central Communication Channel that stores, filters, and distributes messages by round. The LLM-based Communicate module decides whether to send a message and, if so, selects one of six types: information, question, individual strategy, collective strategy, establish agreement, or discuss/evaluate. All messages are logged and stored in LTM so that agents can later recall what others said and what they themselves committed to.

Formal coordination is supported by an Agreement System layered on top of communication. When communication is enabled, agents can propose, accept, or decline explicit agreements using a standardized message format. (e.g., “propose: AG001: No agent should harvest the last apple from trees with only one or two apples remaining”). These interactions are tracked by an AgreementManager that maintains the lifecycle of each agreement (proposed, active, declined), synchronizes active agreements into STM, and records all actions in LTM with dedicated metadata. Active agreements are included as structured context in all cognitive-module prompts, so that perception, planning, action selection, communication, and reflection all condition on the current set of commitments.

This architecture yields LLM-based agents that (i) integrate perception, planning, action, reflection, and communication in a unified cognitive loop, (ii) exploit rich memory structures to reason over past experiences and interactions, and (iii) use compact messages and formal agreements to coordinate strategies and sustain cooperative behavior in the grid-world social dilemma.

1.3.2 Reflection Module

To promote stable cross-episode learning and enable agents to extract generalizable principles beyond a single rollout, each LLM-based agent performs a Reflection Module between scenarios. This module consolidates experience from the preceding episode and generates high-level insights about strategy, environmental dynamics, coordination patterns, and sustainability mechanisms.

Reflection operates by selecting a set of 20 structured questions—10 fixed and 10 dynamically generated—which the agent answers using its long-term memory, spatial memory, and the full log of perceptions, actions, and communications. Fixed questions promote broad strategic introspection, while dynamic questions are automatically derived from salient events in the episode (e.g., resource fluctuations, violations of agreements, regrowth patterns, or communication-behavior mismatches). This ensures both stability and adaptivity: agents reflect on universal principles while also reasoning about environment-specific causal structure.

During the reflection step, the agent:

- Summarizes recent experiences into memory chunks.
- Selects 20 questions: 10 fixed high-level questions + 10 dynamic episode-specific questions.
- Produces concise answers that are stored in long-term memory and used as priors for the next scenario’s planning module.

This process provides a mechanism for transfer of knowledge across scenarios, supports more coherent cooperative behavior over long horizons, and improves the agent’s ability to detect patterns of resource depletion, environmental dynamics, and social coordination.

Fixed Reflection Questions (Global, Episode-Independent)

1. What strategies have I found most effective for maximizing my reward in this environment?
2. What patterns have I observed about resource availability, regeneration, or depletion?
3. What have I learned about interacting with other agents (cooperation, competition, avoidance)?
4. What situations or actions have consistently led to negative outcomes or reduced rewards?
5. What spatial patterns or locations have been particularly valuable or dangerous?
6. What have I learned about the timing or sequencing of actions for optimal outcomes?
7. How does the environment change over time, and how should I adapt my strategy?
8. What trade-offs have I identified between different objectives or strategies?
9. What communication strategies or information-sharing approaches have been most valuable?
10. What general principles or lessons have I learned that might apply in similar situations?

In addition to the 10 fixed high-level questions, the Reflection Module generates episode-specific dynamic questions derived from salient events in the just-completed scenario. These questions are produced automatically. Below we present an example set of dynamic questions taken from one representative simulation, illustrating how the module adapts its introspection to the environmental and social dynamics observed in that episode:

1. How do the foraging behaviors of myself and Agente2, including single-apple harvesting and movement patterns, influence long-term apple and grass regeneration in key trees?
2. How does the local distribution of apples and grass around tree 5 determine which tiles regrow apples or lose grass, and what triggers these transitions?
3. How do apple and grass tiles around tree 6 change after harvesting events by any agent?
4. How effective has agent communication (e.g., moratoriums, risk warnings) been in shaping harvesting behavior and supporting cooperative resource management?
5. What is the current sustainability outlook for harvesting apples from tree 5, given its recent regrowth and grass conditions?
6. What relationship exists between local apple density (L2 radius) and the regrowth or disappearance of apples/grass around tree 6?
7. Are there implicit or explicit cooperative agreements (e.g., respecting moratoriums), and how do they influence agent behavior and sustainability?
8. Under what conditions do grass tiles disappear or regrow apples, and how do these depend on nearby apple distributions?
9. What trade-offs exist between immediate collection and long-term regeneration in recent harvesting variations across agents and trees?
10. How does apple and grass regrowth in tree 5 depend on apple counts and positions, and what patterns were observed in the latest steps?

2 Well-being Variables Details

This section describes the well-being variables employed to measure cooperative resilience. For the well-being variables, the figures in this section present the mean values across 5 episodes, with blue representing performance and orange indicating the reference. Shaded areas illustrate the standard deviations. Red dashed lines mark the onset of disruptive events.

2.1 Apples alive *per capita*

This variable evaluates the number of living apples in the environment at each step. This measure is directly related to the availability of resources for the community, thus reflecting the well-being of the agents. Therefore, a higher number of apples *per capita* indicates better future access to resources for the agents with out communication.

Figure S3 reports the trajectories of this variable over 250 time steps for human agents across the nine scenarios. The corresponding results with communication over the same horizon are shown in Figure S4. Long-horizon experiments with 1000 time steps, without and with communication, are presented in Figure S5 and Figure S6, respectively.

Figure S7 shows the corresponding trajectories for LLM agents in the 250-timestep setting without communication, and Figure S8 presents the communication variant.

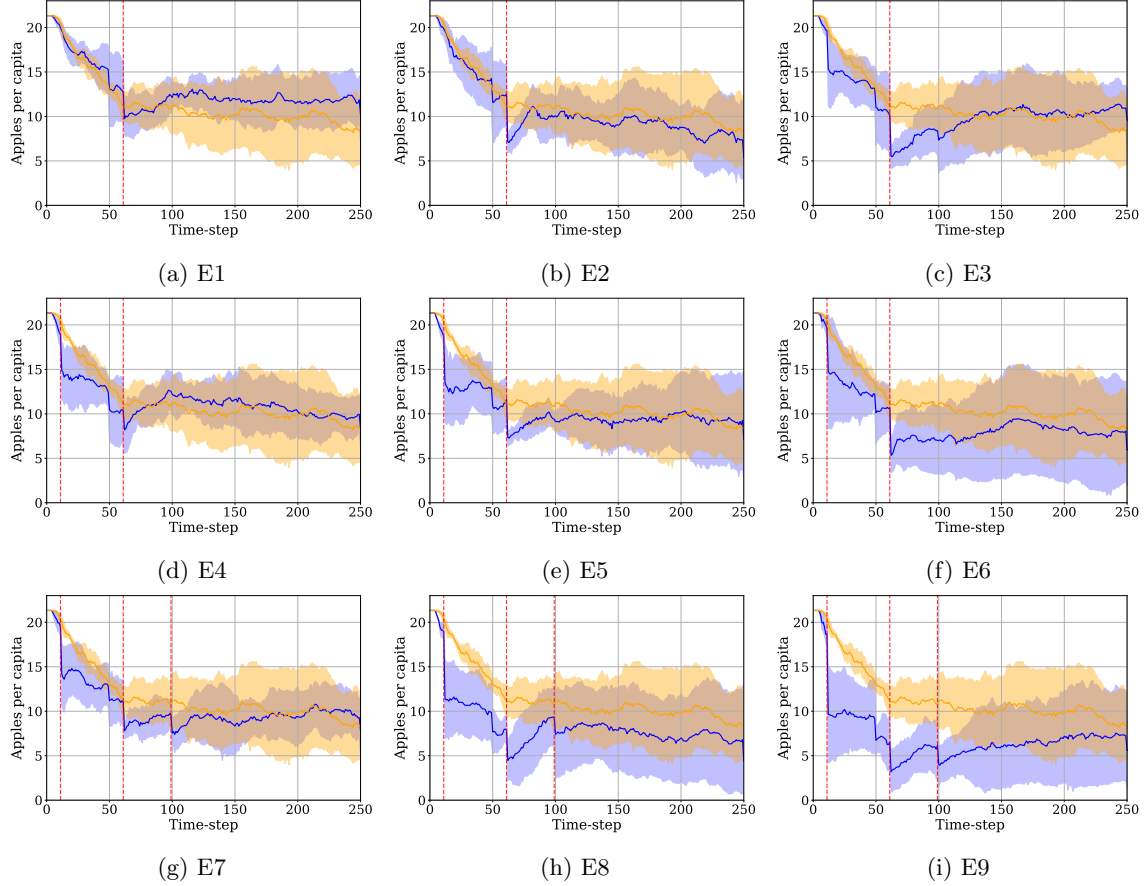


Figure S3: Performance and reference curves of *apples alive per capita* for human agents without communication in the 250-timestep setting.

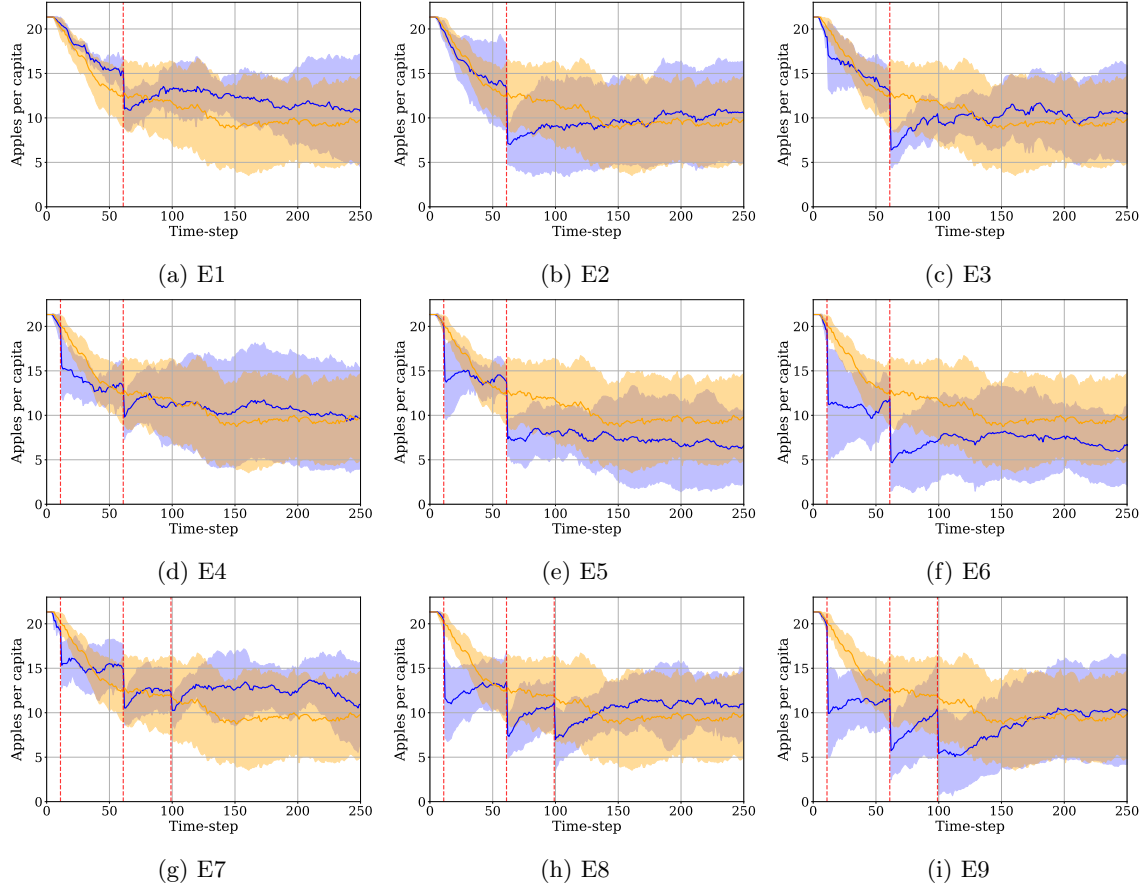


Figure S4: Performance and reference curves of *apples alive per capita* for human agents with communication in the 250-timestep setting.

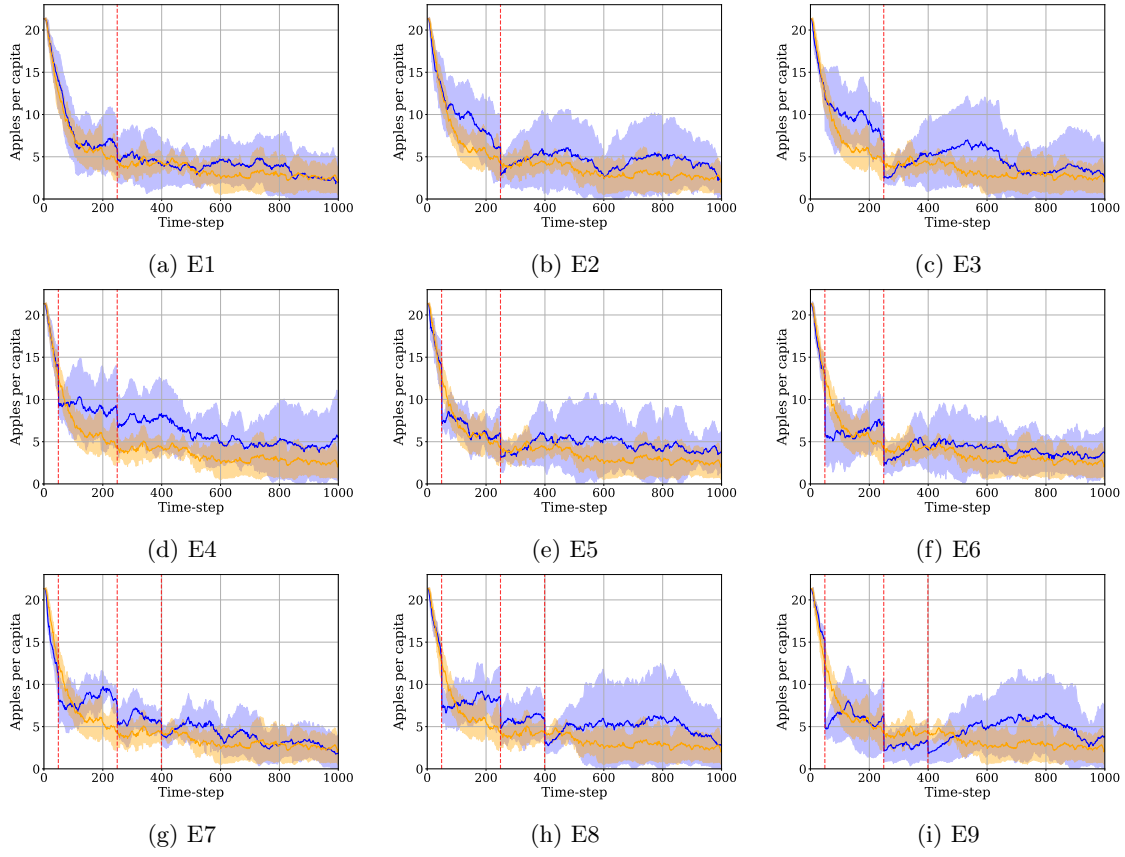


Figure S5: Performance and reference curves of *apples alive per capita* for human agents without communication in the 1000-timestep setting.

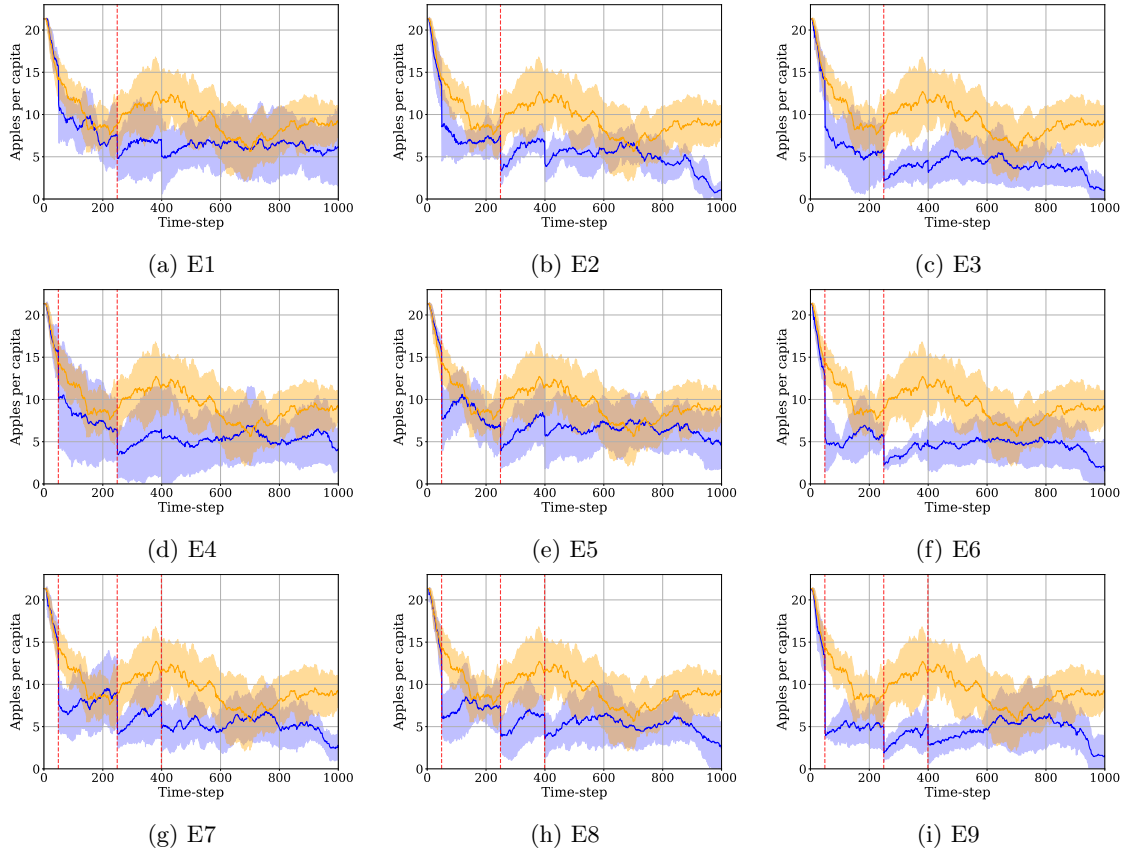


Figure S6: Performance and reference curves of *apples alive per capita* for human agents with communication in the 1000-timestep setting.

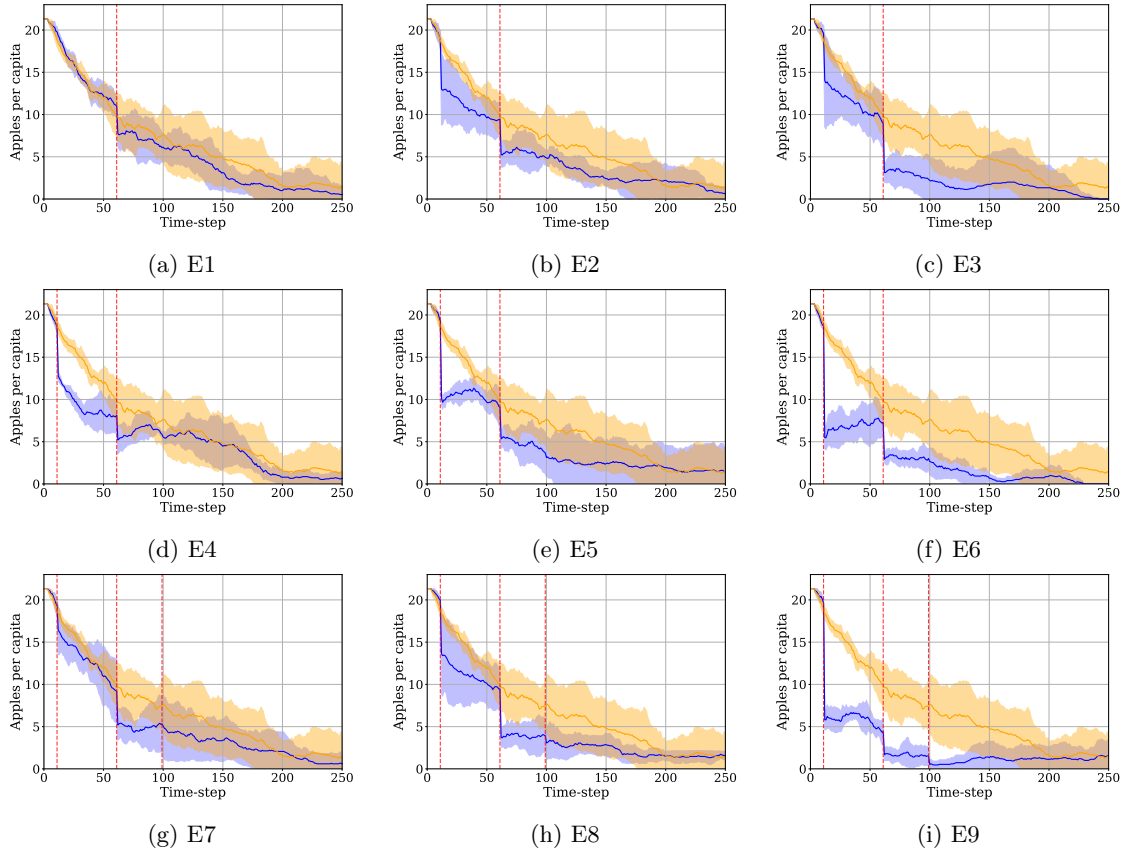


Figure S7: Performance and reference curves of *apples alive per capita* for LLM agents without communication in the 250-timestep setting.

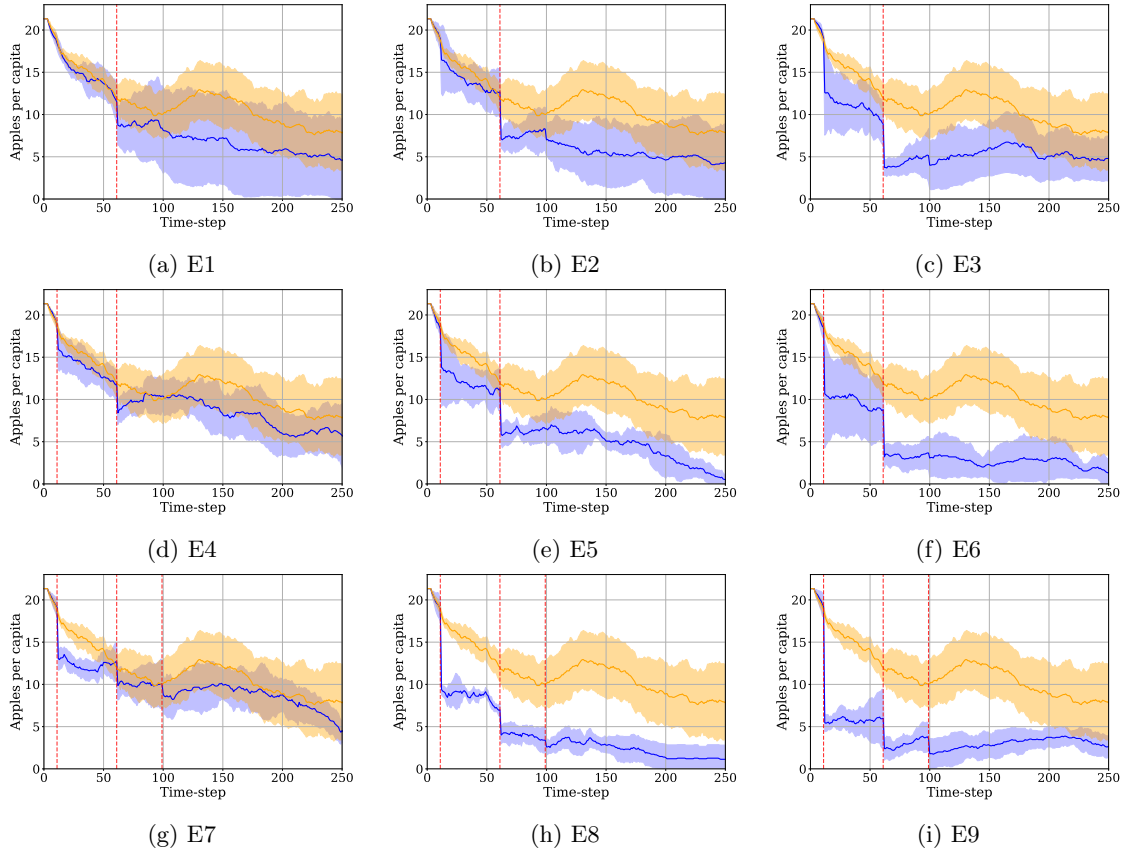


Figure S8: Performance and reference curves of *apples alive per capita* for LLM agents with communication in the 250-timestep setting.

2.2 Trees alive *per capita*

Similar to apples alive *per capita* this variable evaluates the number of living tree in the environment at each step. This measure is also directly related to the availability of resources for the community. Therefore, a higher number of tree *per capita* indicates better future access to resources for the agents. Additionally, the trees also highlight an important aspect related to the recovery systems. When a tree disappears, it never reappears, resulting in a significant decrease in resource (apples) availability.

Figure S9 reports the performance and reference trajectories of this variable for human agents without communication in the 250-timestep setting, while Figure S10 presents the corresponding results with communication. The long-horizon experiments with 1000 time steps are shown in Figure S11 (without communication) and Figure S12 (with communication). Figure S13 and Figure S14 display the analogous results for LLM agents without and with communication, respectively.

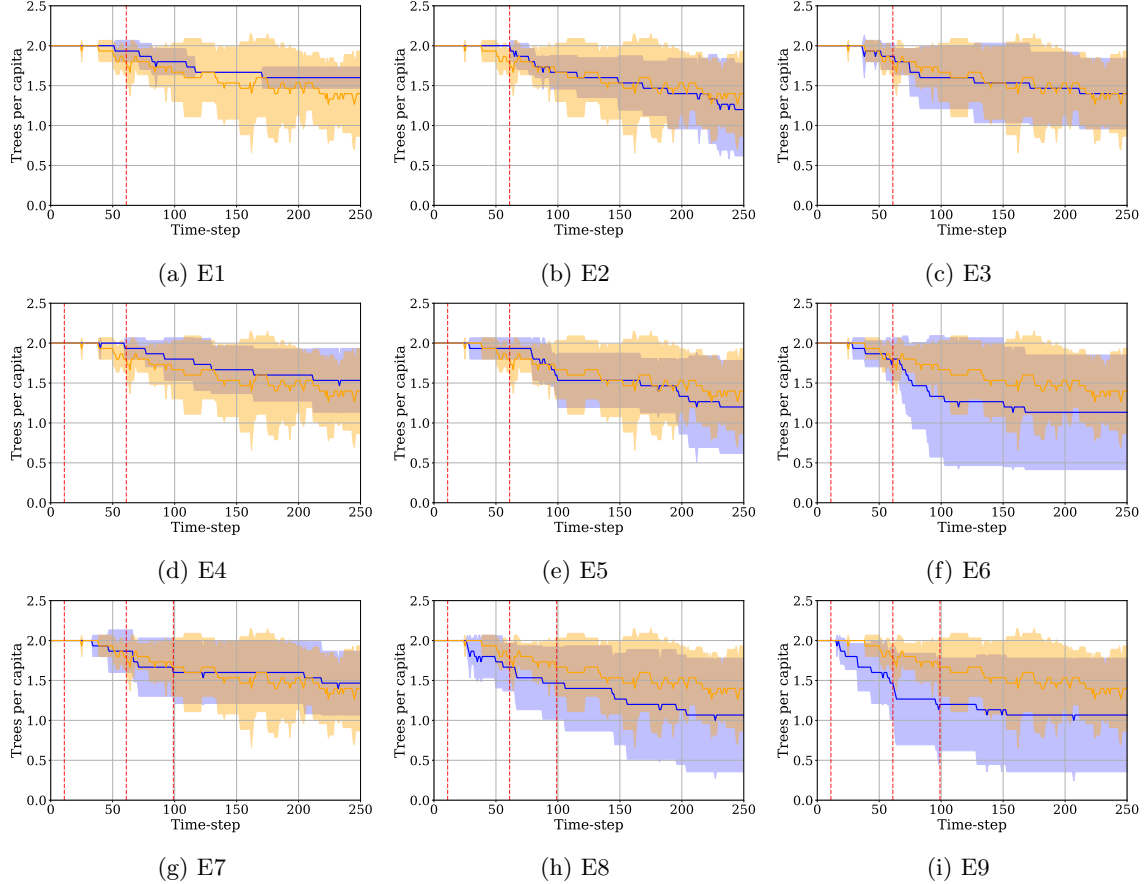


Figure S9: Performance and reference curves of *trees alive per capita* for human agents without communication in the 250-timestep setting.

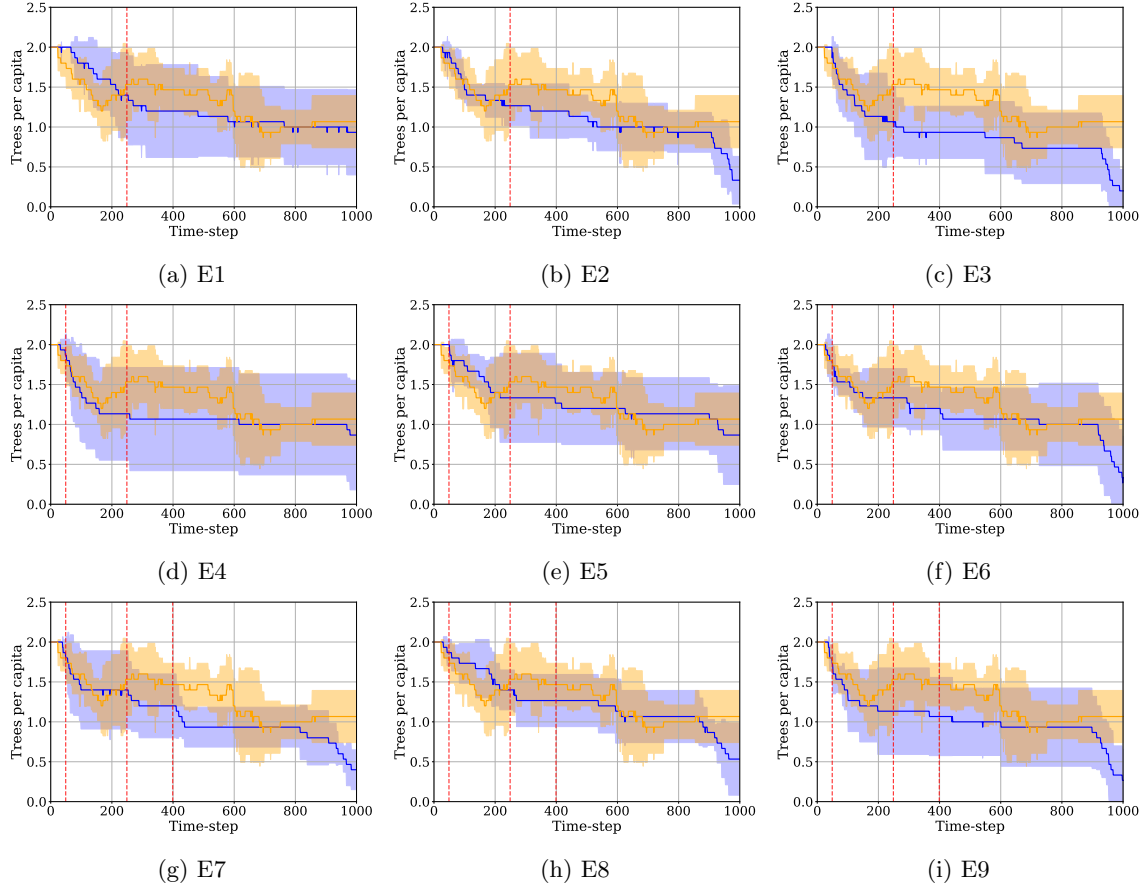


Figure S10: Performance and reference curves of *trees alive per capita* for human agents with communication in the 250-timestep setting.

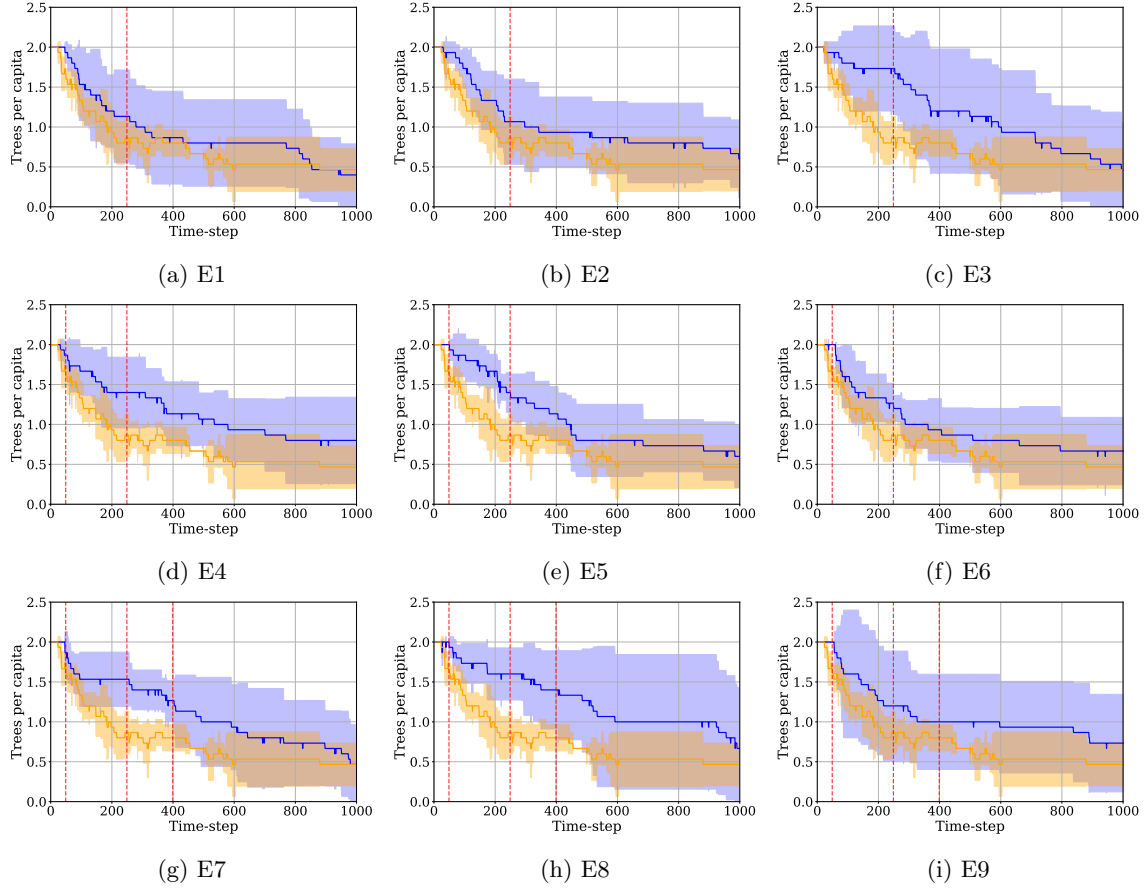


Figure S11: Performance and reference curves of *trees alive per capita* for human agents without communication in the 1000-timestep setting.

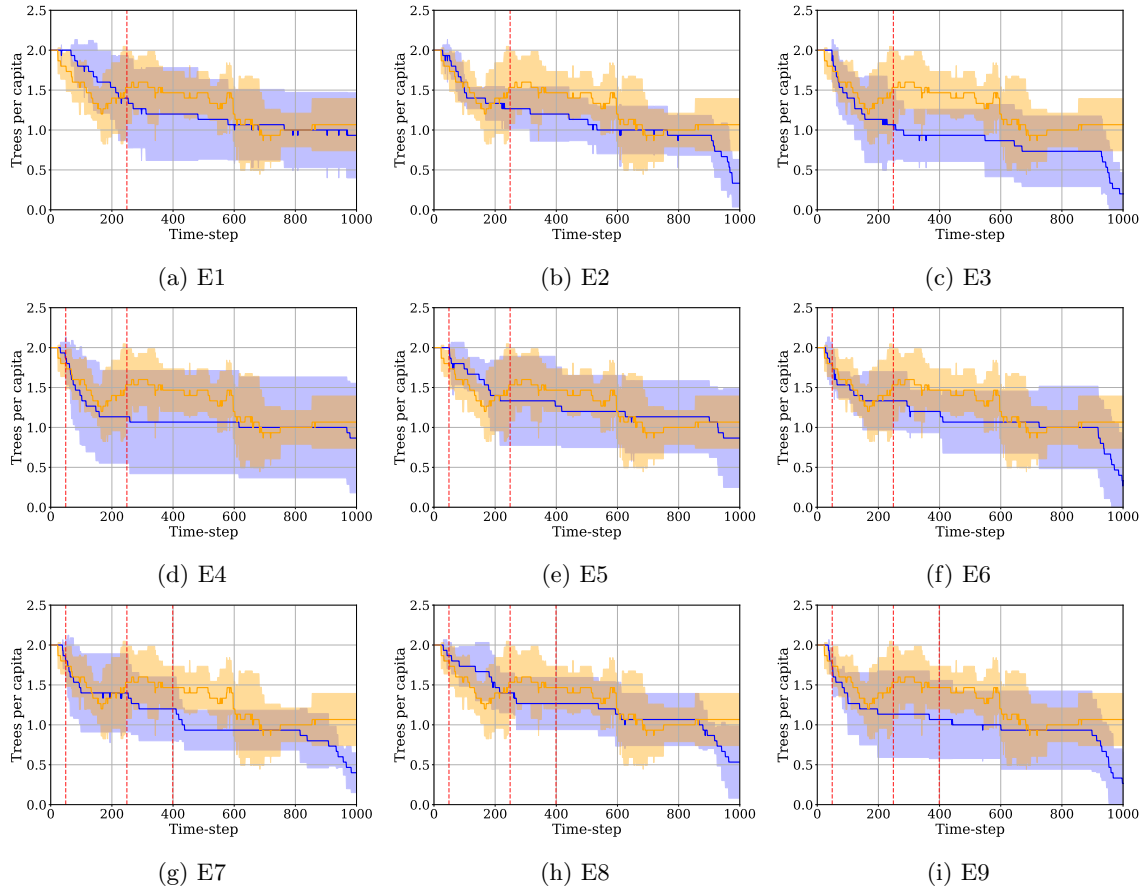


Figure S12: Performance and reference curves of *trees alive per capita* for human agents with communication in the 1000-timestep setting.

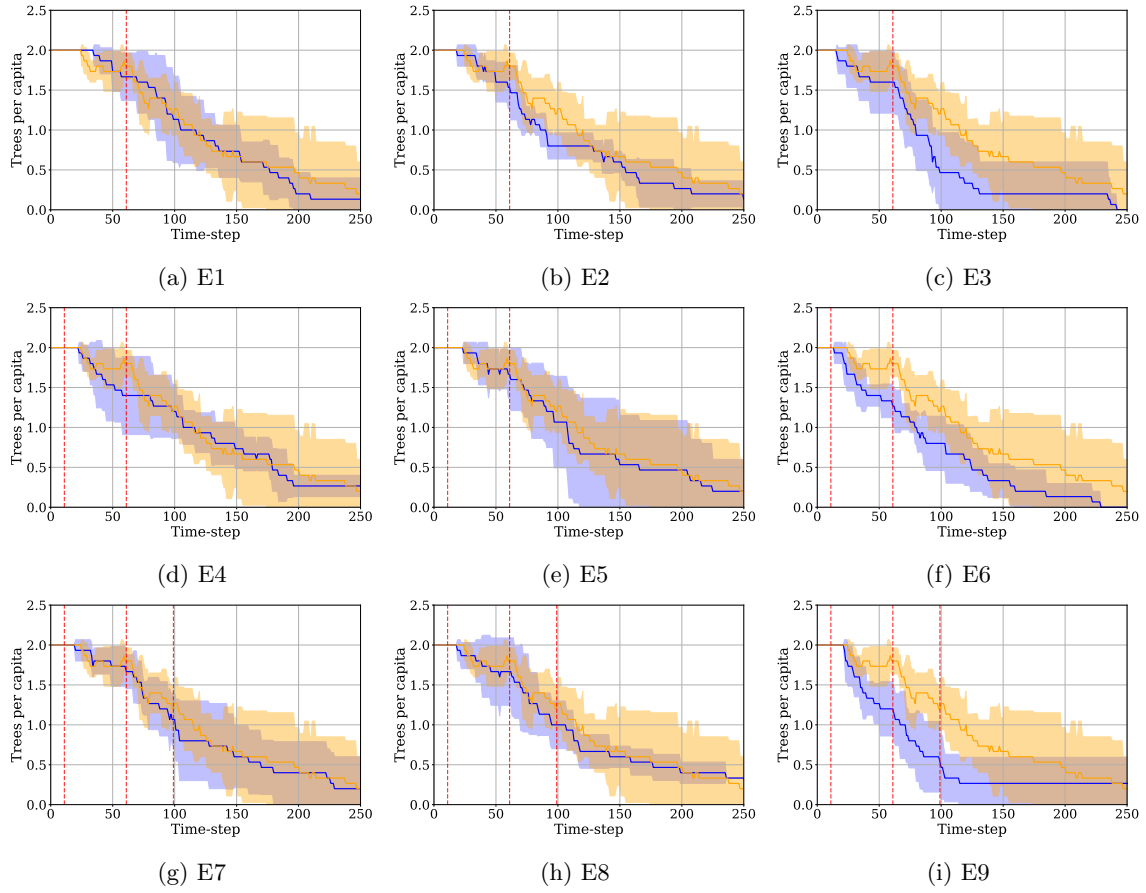


Figure S13: Performance and reference curves of *trees alive per capita* for LLM agents without communication in the 250-timestep setting.

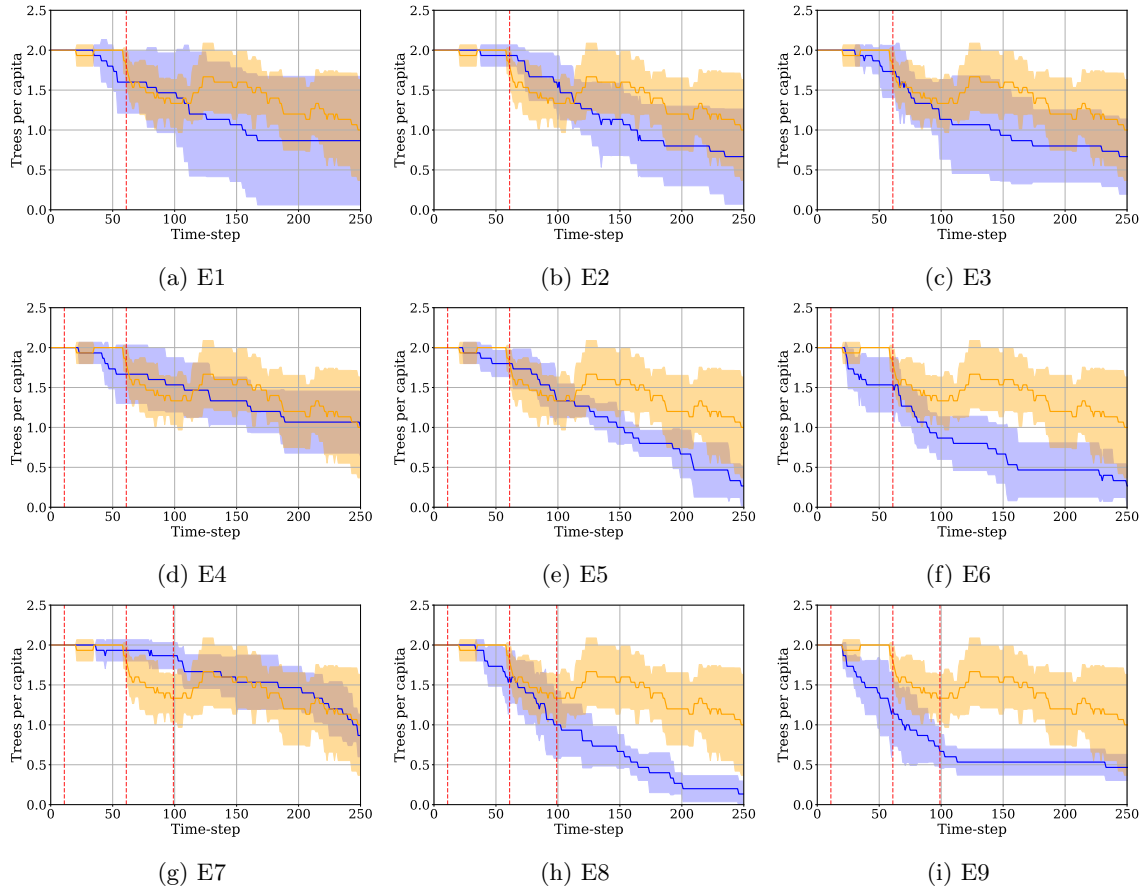


Figure S14: Performance and reference curves of *trees alive per capita* for LLM agents with communication in the 250-timestep setting.

2.3 Cumulative Gini Equality Index

This indicator builds on the Gini coefficient, widely used in economics to quantify disparities in resource distributions, and repurposes it to capture how evenly rewards are shared among agents. Gini-based measures have been used in multi-agent research to study equity in sequential social dilemmas [S2], while [S3] employed the inverse Gini coefficient as a learning signal in RL settings related to the Melting Pot framework. This indicator is included as part of the resilience methodology proposed in [S4].

Figure S15 presents the trajectories of the Gini-based fairness indicator for human agents without communication in the 250-timestep setting, while Figure S16 shows the corresponding results with communication. The long-horizon variants with 1000 time steps are reported in Figure S17 (no communication) and Figure S18 (communication). Figure S19 and Figure S20 provide the analogous curves for LLM agents, without and with communication, respectively.

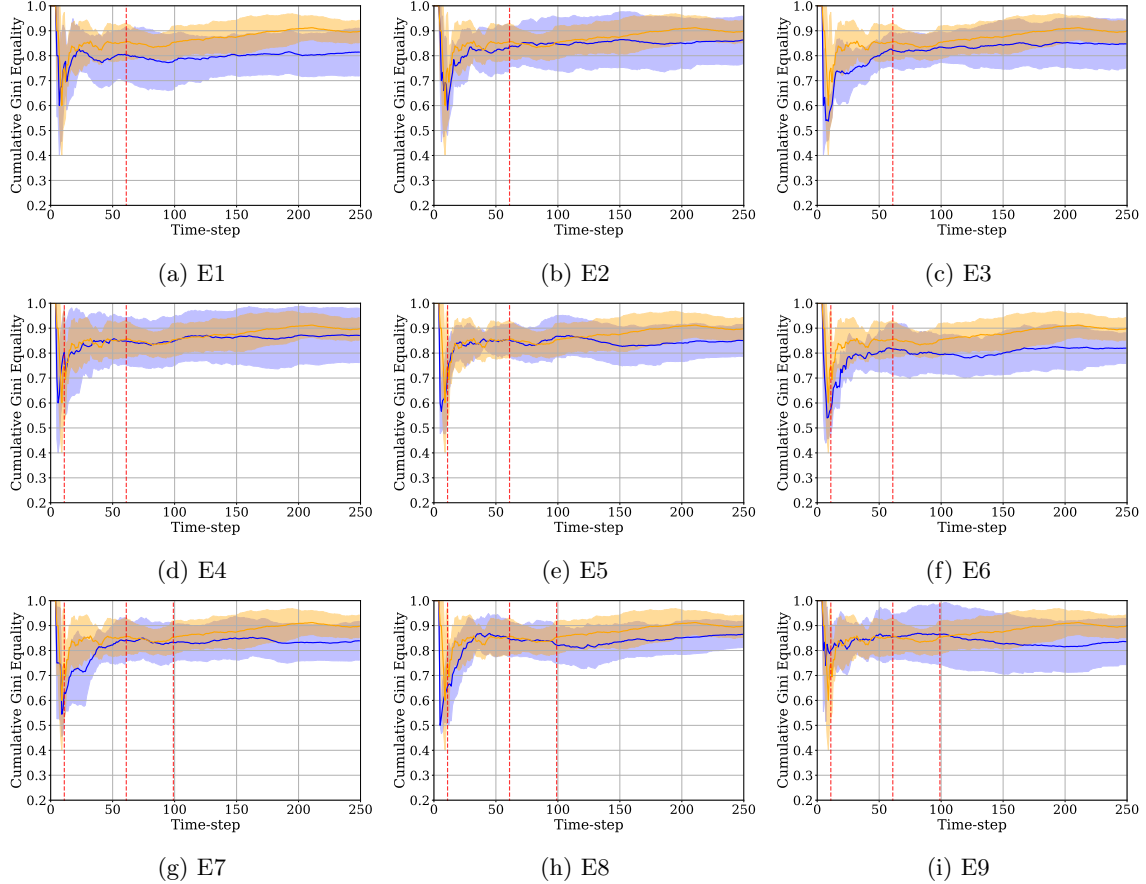


Figure S15: Performance and reference curves of Cumulative Gini Equality Index for human agents without communication in the 250-timestep setting.

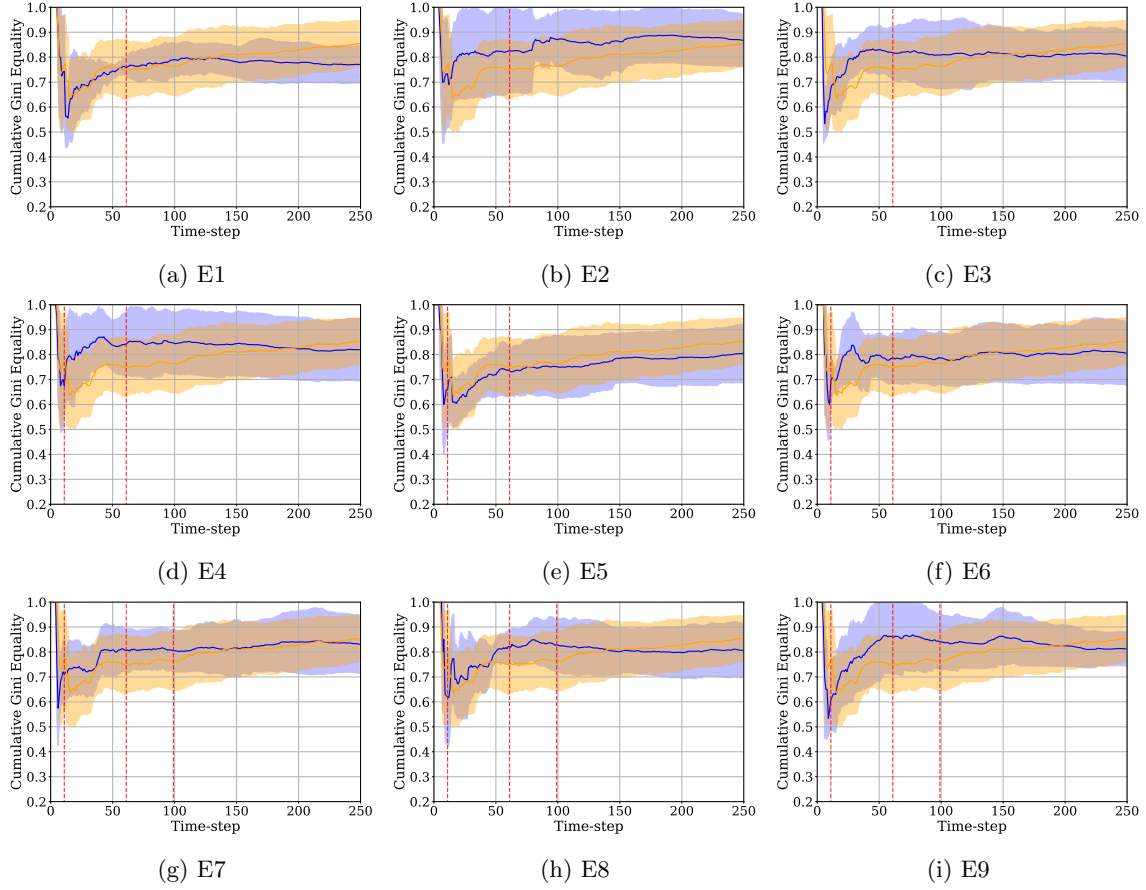


Figure S16: Performance and reference curves of Cumulative Gini Equality Index for human agents with communication in the 250-timestep setting.

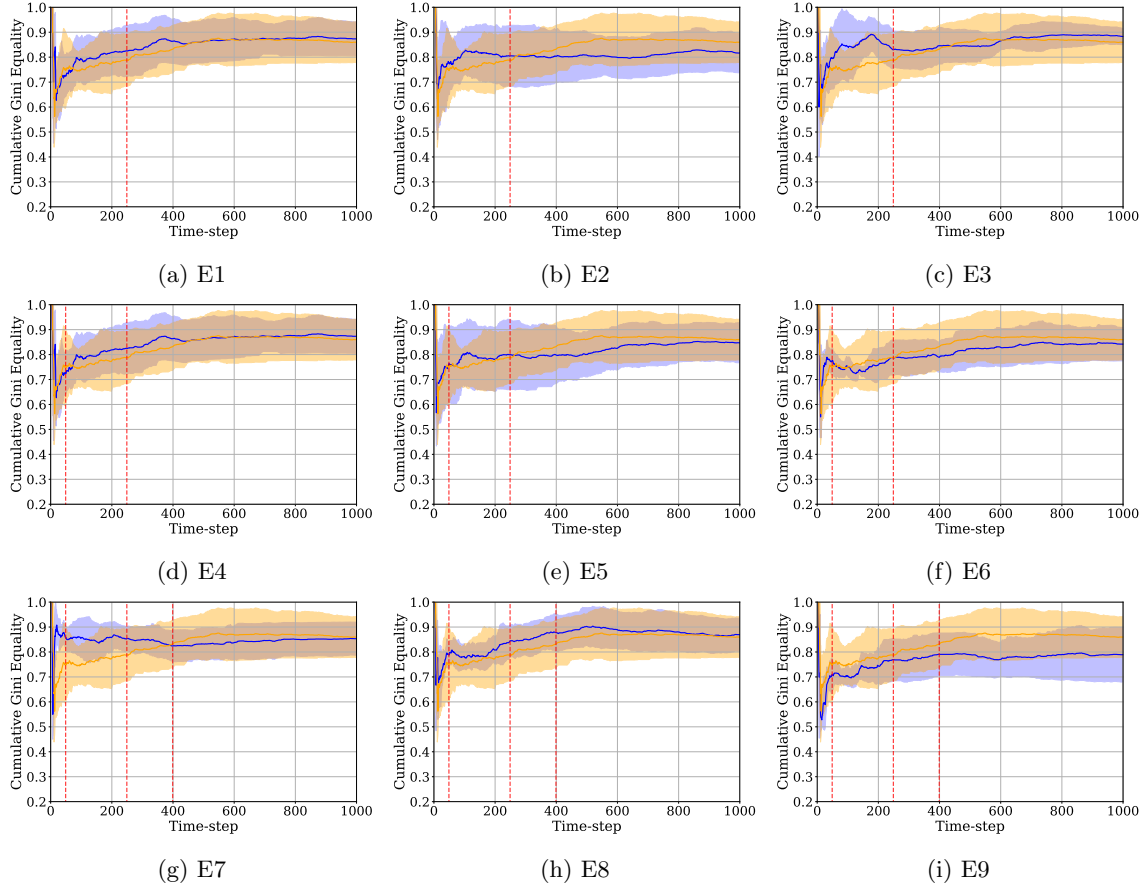


Figure S17: Performance and reference curves of Cumulative Gini Equality Index for human agents without communication in the 1000-timestep setting.

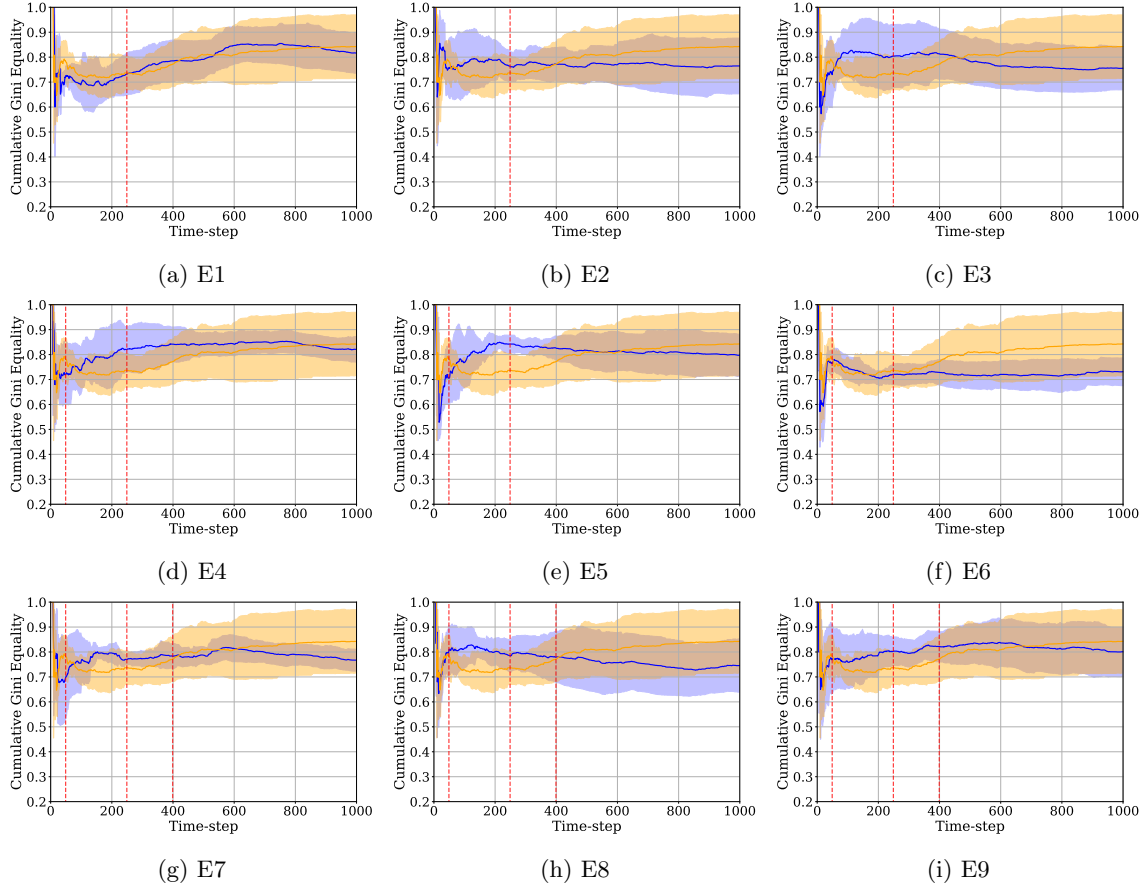


Figure S18: Performance and reference curves of Cumulative Gini Equality Index for human agents with communication in the 1000-timestep setting.

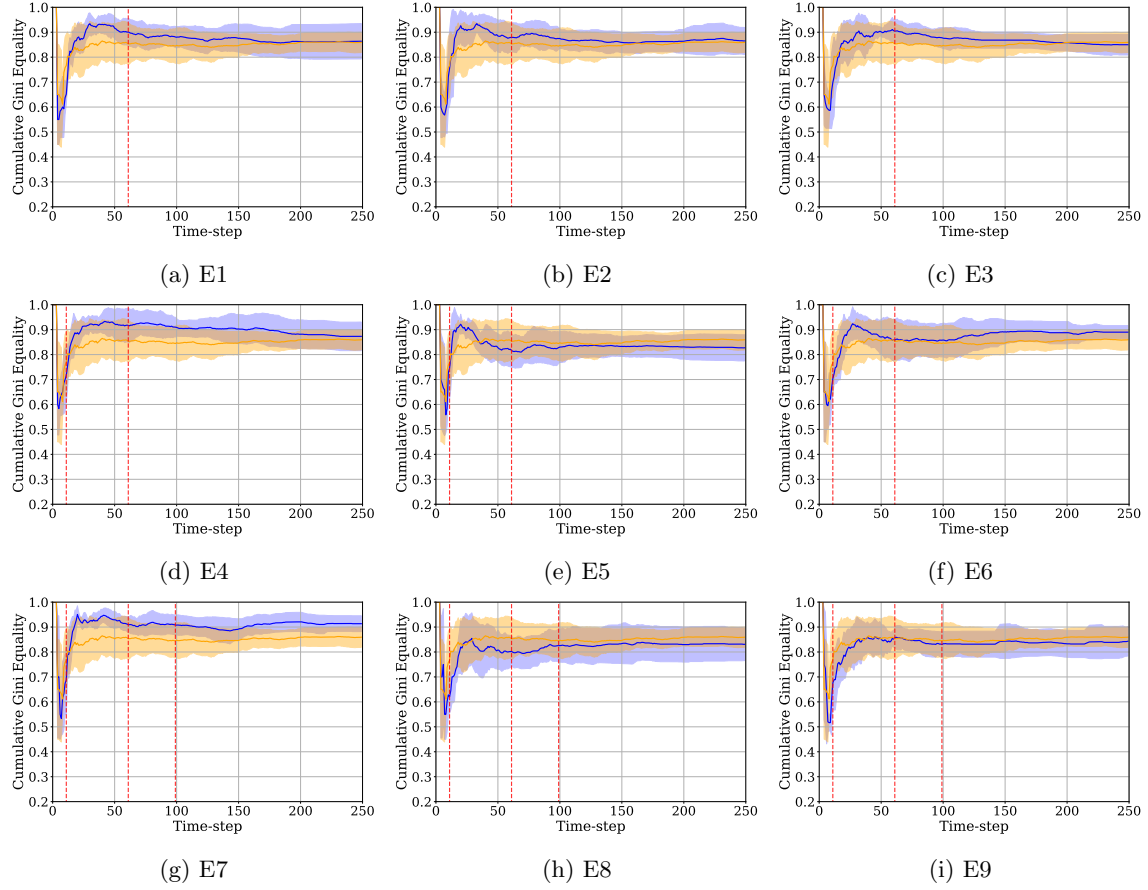


Figure S19: Performance and reference curves of Cumulative Gini Equality Index for LLM agents without communication in the 250-timestep setting.

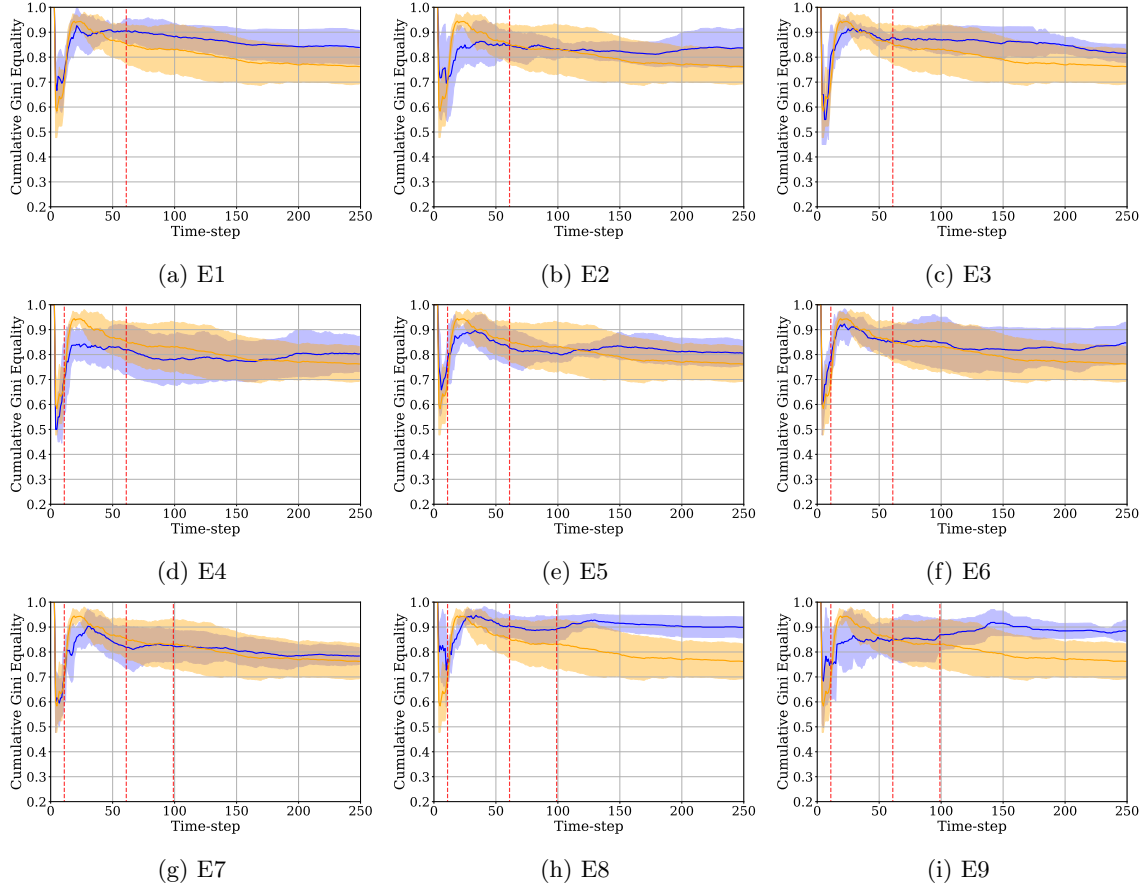


Figure S20: Performance and reference curves of Cumulative Gini Equality Index for LLM agents with communication in the 250-timestep setting.

2.4 Collective Satiation Index

We employ a *collective satiation index* that tracks how recently and how often agents receive rewards, converting each agent’s discounted reward history into a satiation score in $[0,1]$. These individual scores are then aggregated via a harmonic mean, so the index is high only when all agents eat regularly and drops sharply if any agent remains unsatiated. This indicator is adapted from the resilience methodology proposed in [S4].

Figure S21 presents the trajectories of the collective satiation index for human agents without communication in the 250-timestep setting, while Figure S22 shows the corresponding results with communication. The long-horizon variants with 1000 time steps are reported in Figure S23 (no communication) and Figure S24 (communication). Figure S25 and Figure S26 provide the analogous curves for LLM agents, without and with communication, respectively.

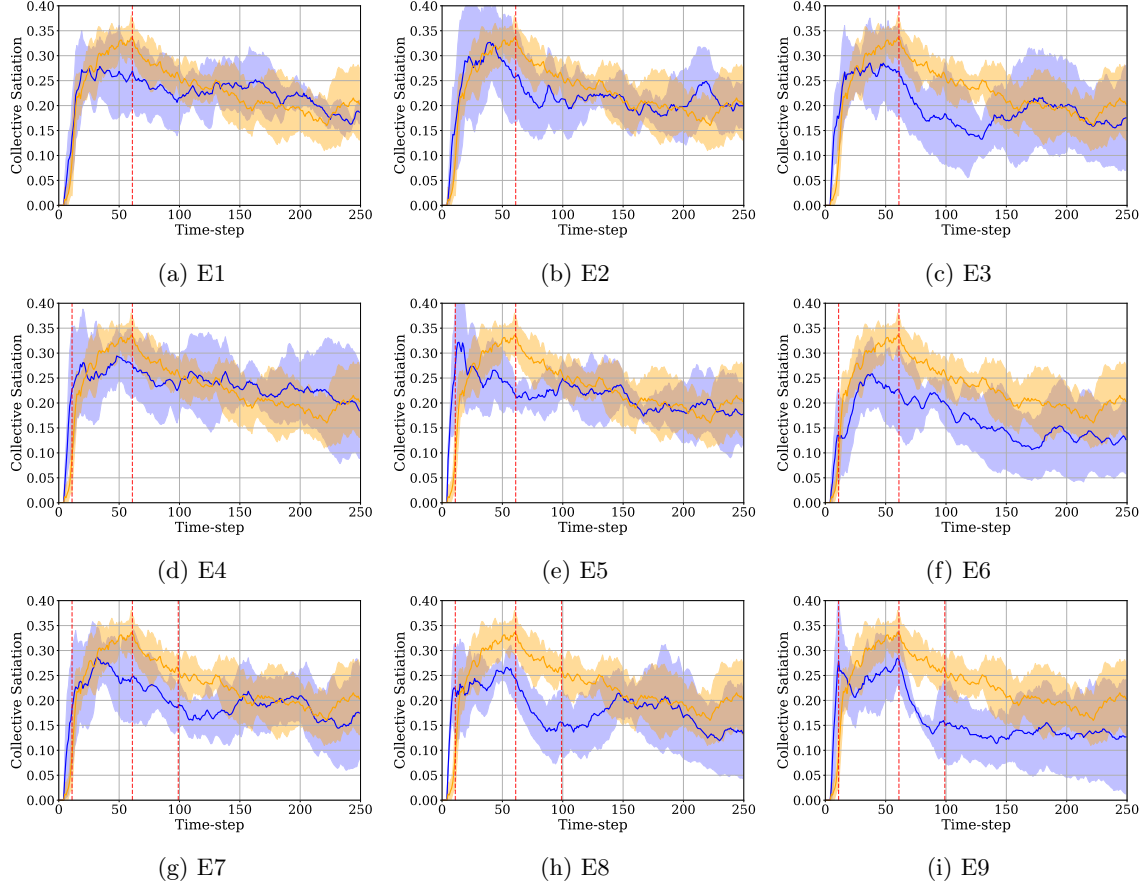


Figure S21: Performance and reference curves of Collective Satiation Index for human agents without communication in the 250-timestep setting.

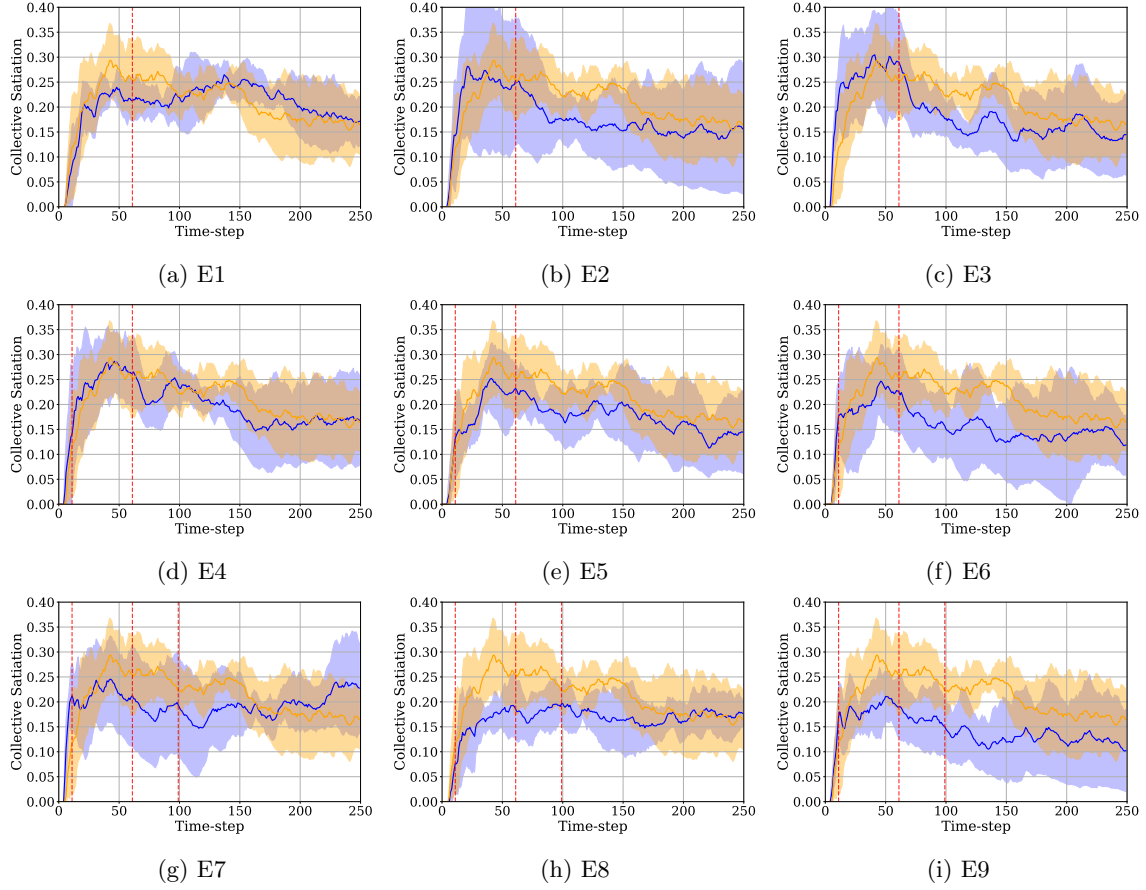


Figure S22: Performance and reference curves of Collective Satiation Index for human agents with communication in the 250-timestep setting.

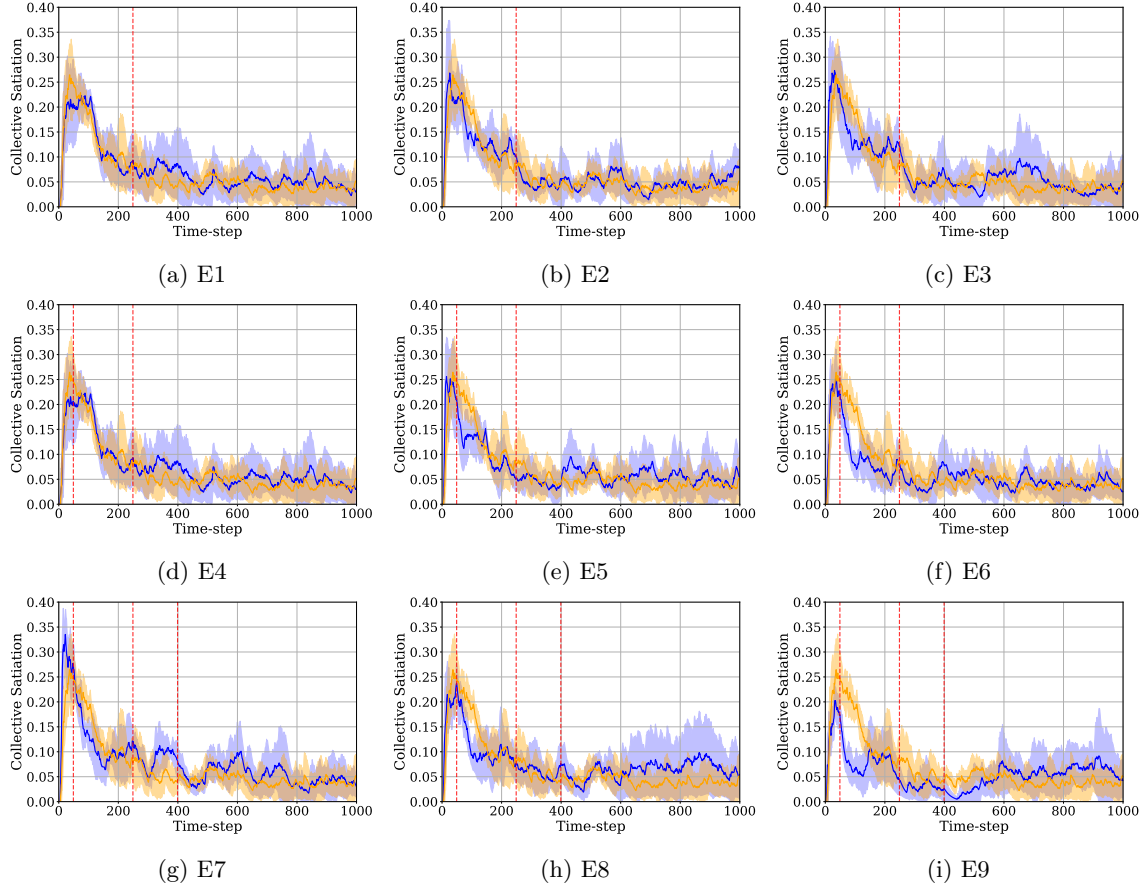


Figure S23: Performance and reference curves of Collective Satiation Index for human agents without communication in the 1000-timestep setting.

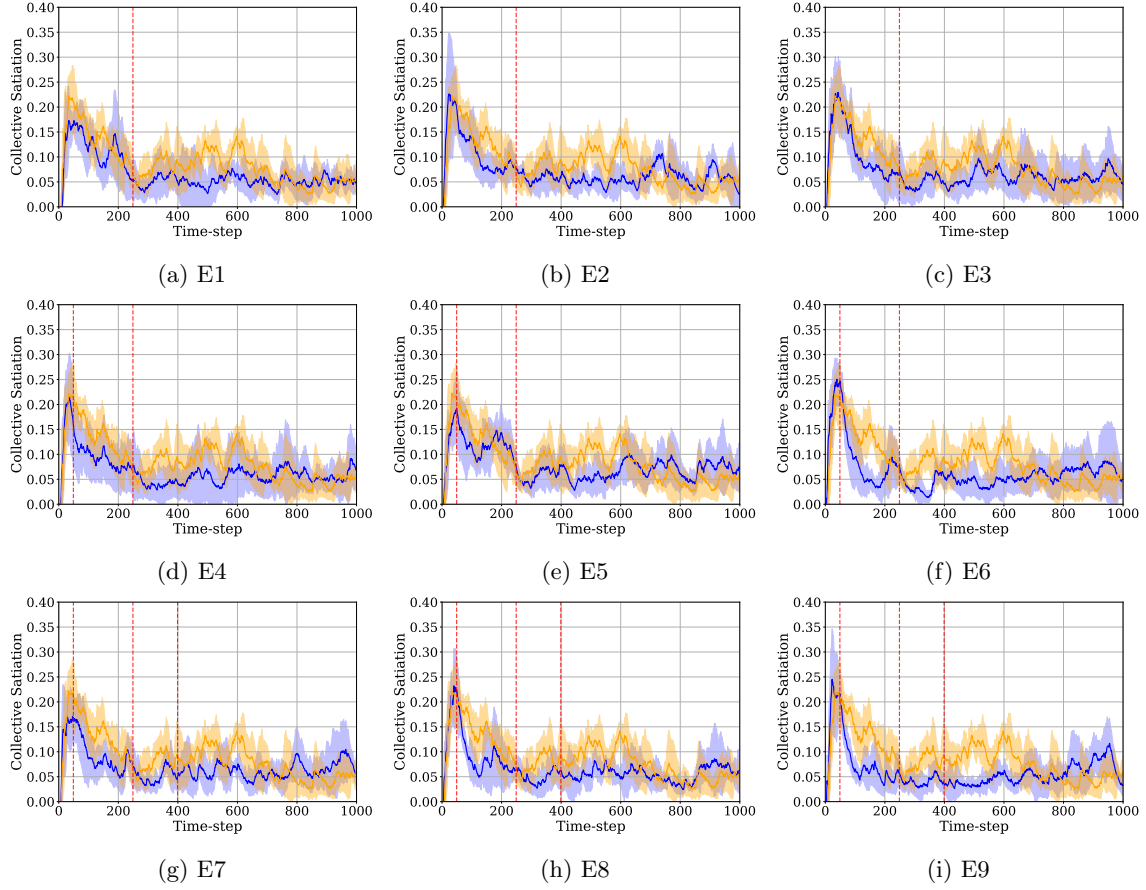


Figure S24: Performance and reference curves of Collective Satiation Index for human agents with communication in the 1000-timestep setting.

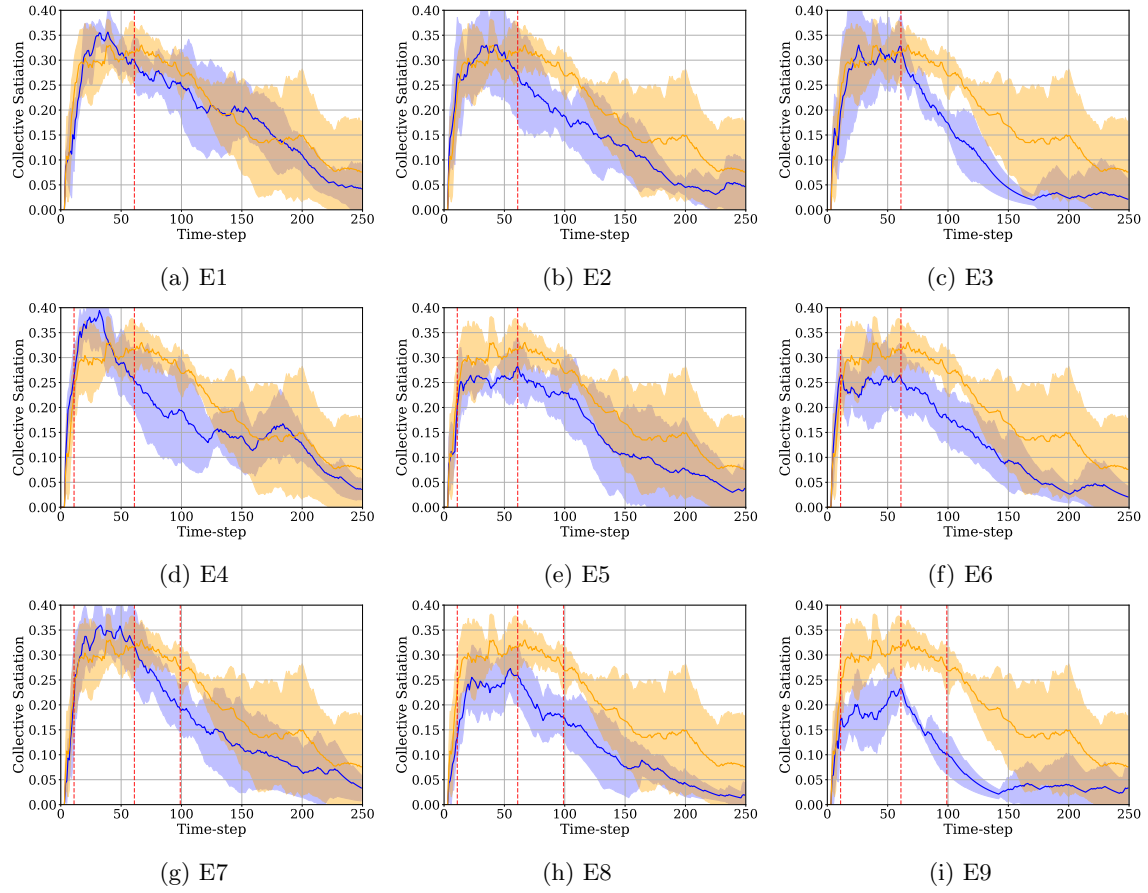


Figure S25: Performance and reference curves of Collective Satiation Index for LLM agents without communication in the 250-timestep setting.

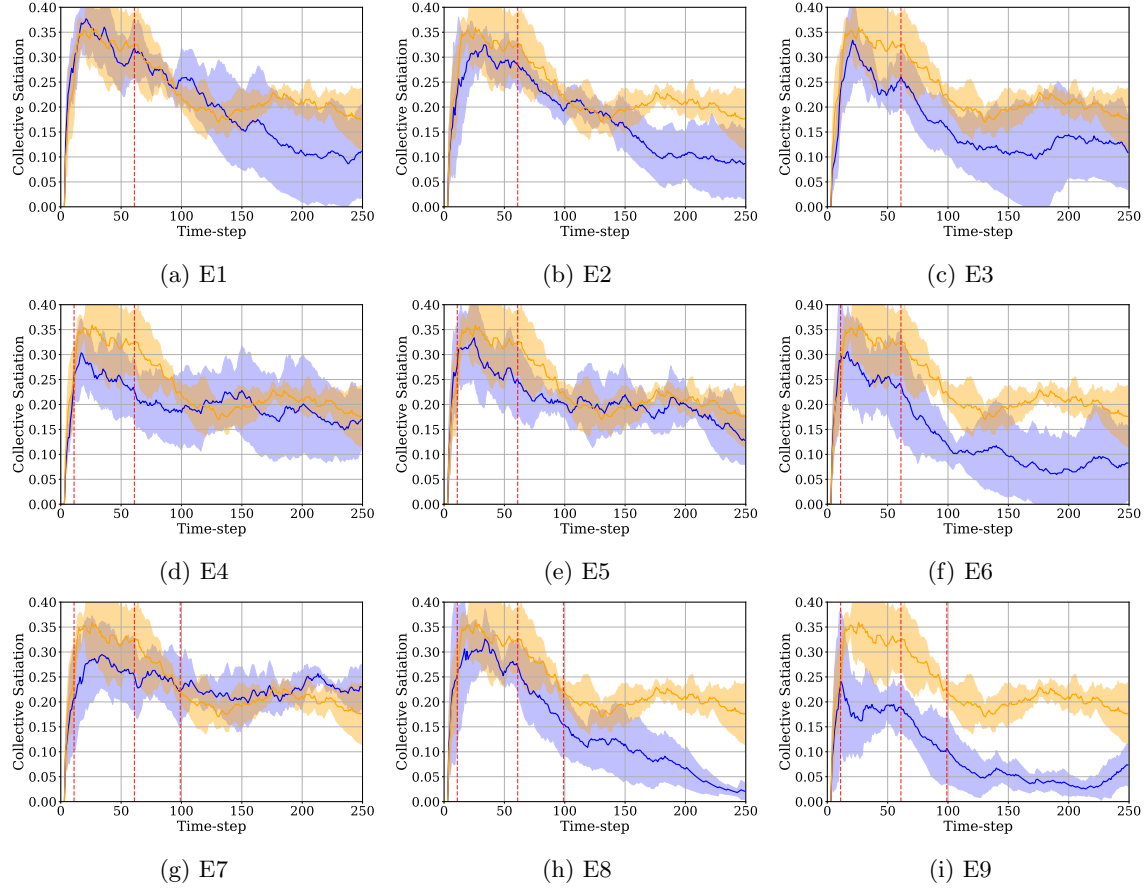


Figure S26: Performance and reference curves of Collective Satiation Index for LLM agents with communication in the 250-timestep setting.

3 Shared Baseline

This appendix expands the methodology behind the shared baselines used to enable cross-group comparability in cooperative resilience. Building on the original approach of [S4], where resilience was evaluated relative to each system’s own non-disrupted performance, the present study requires a common reference because humans and LLM agents exhibit systematically different behaviors even in non-disrupted settings. To establish a unified benchmark, we computed empirical baselines for each metric using all non-disrupted runs across the experimental set: humans and LLMs (250-timestep, with/without communication) and the other set: humans (1000-timestep, with/without communication). For the 250-timestep experiments, we considered the four well-being indicators.

In the first set humans and LLMs (250-timestep, with/without communication) for each indicator and for each of the four non-disrupted reference conditions (Hum., Hum. + Comm., LLM, LLM + Comm.), we computed the median value across the entire temporal horizon (250 time steps). This yields four condition-specific medians per indicator. The shared baseline for a given indicator is then defined as the median of these four medians. Repeating this procedure for all four well-being indicators results in a set of consistent, condition-agnostic baseline values that are used as reference lines in the corresponding plots. This procedure is illustrated in Figure S27, which displays the four well-being indicators together with their corresponding shared baseline lines. As shown, the resulting baseline is not biased toward either the best- or worst-performing baseline condition, but instead provides a balanced reference across all non-disrupted behaviors. Table S1 summarizes, for each indicator, the condition-specific medians and the final median-of-medians value used as the global baseline.

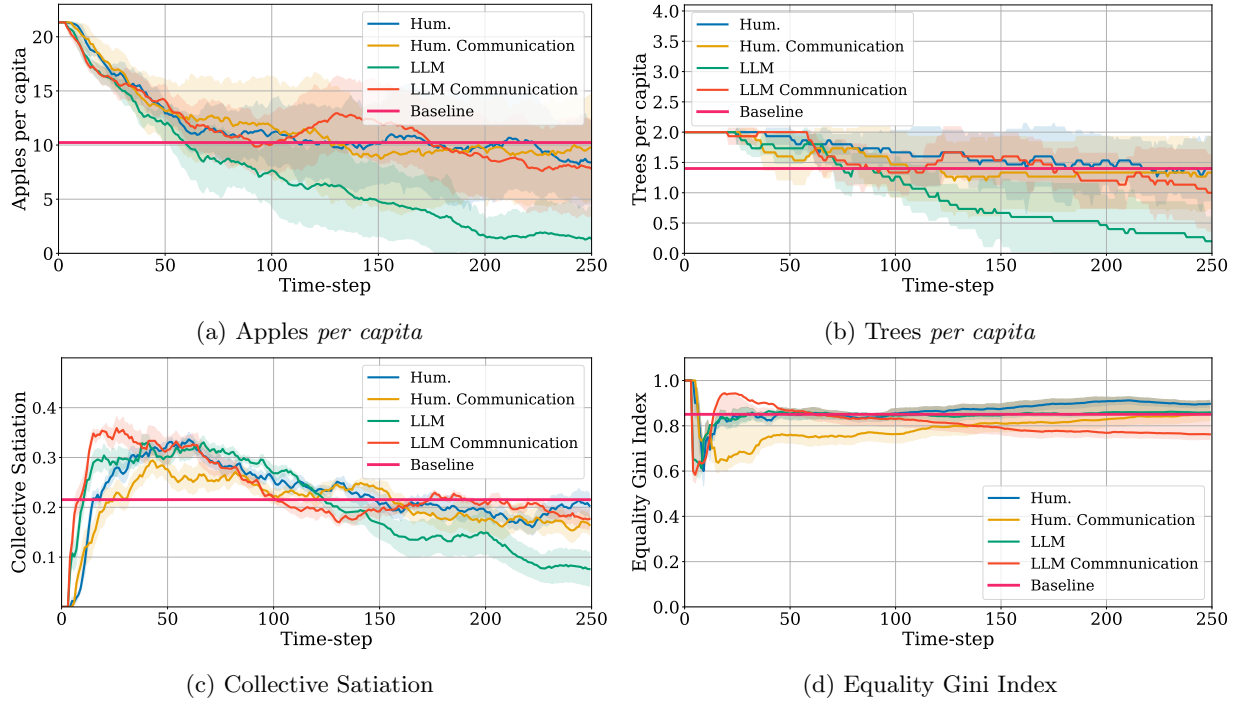


Figure S27: Baseline construction for the four well-being indicators from the first set of non-disrupted conditions (Hum., Hum. + Comm., LLM, LLM + Comm.), using a median-of-medians aggregation.

For the second set of experiments, involving humans in the extended 1000-timestep environment (with and without communication), we apply the same median-of-medians procedure, now over the longer temporal horizon. For each well-being indicator and for each of the two non-disrupted reference conditions (Hum. 1000, Hum. 1000 + Comm.), we compute the median value across the 1000 time steps, yielding two condition-specific medians per indicator. The long-horizon baseline for a given indicator is then defined as the median of these two medians, providing a shared reference for comparing the two long-horizon human conditions.

This construction is illustrated in Figure S28, which shows the four well-being indicators together with their corresponding long-horizon baseline lines. As these baselines are derived exclusively from the 1000-timestep conditions, they are intended for comparisons between the long-horizon experiments only, and not for direct quantitative comparison with the 250-timestep baselines. Table S2 summarizes, for each indicator, the condition-specific medians and the final median-of-medians value used as the long-horizon global baseline.

Table S1: Baseline Extraction Summary

Metric	Hum.	Hum. + Comm.	LLM	LLM + Comm	Baseline
Apples <i>per capita</i>	10.47	10.00	5.97	11.13	10.23
Trees <i>per capita</i>	1.60	1.33	0.83	1.47	1.40
Collective Satiation	0.22	0.22	0.19	0.21	0.22
Equality Gini Index	0.90	0.84	0.86	0.77	0.85

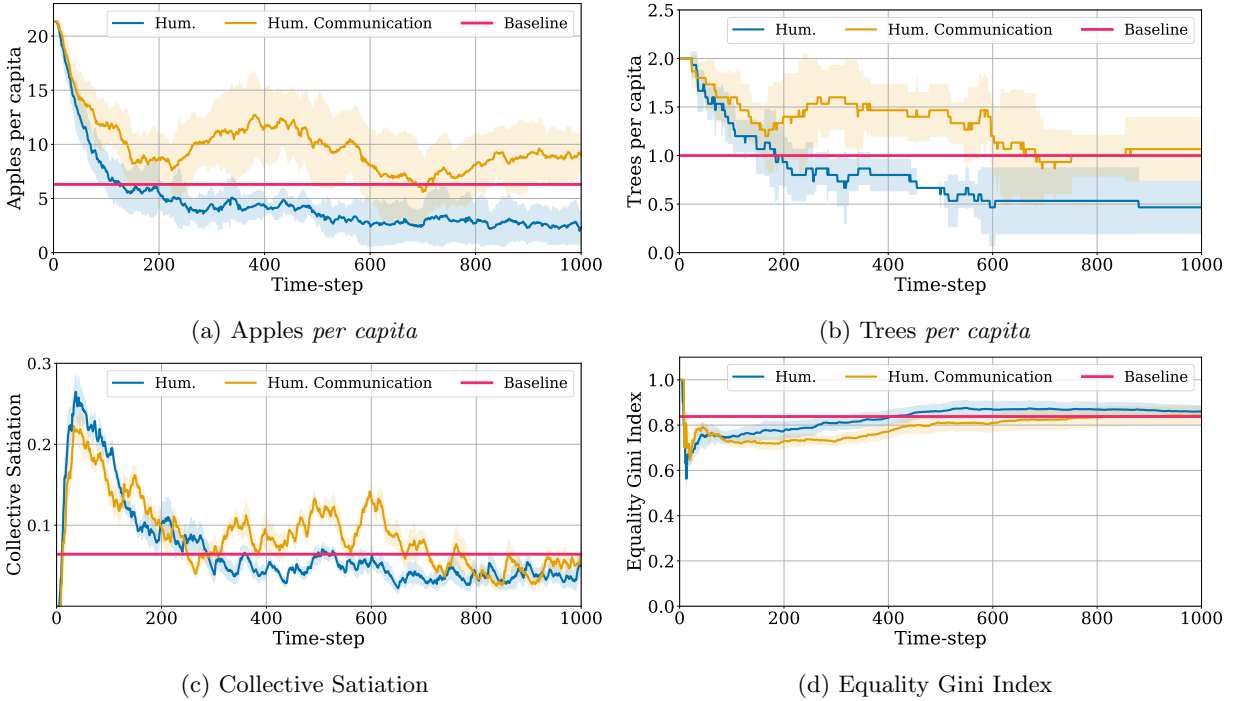


Figure S28: Baseline construction for the four well-being indicators from the second set of non-disrupted conditions (Hum., Hum. + Comm.), using a median-of-medians aggregation.

4 Ethical Considerations

All experiments involving human participants were conducted in accordance with institutional ethical standards and approved protocols. The study received ethical approval from the Institutional Review Board and Ethics Committee of the School of Engineering at the Universidad de los Andes., Bogotá, Colombia. The approved protocol corresponds to **Ethical Approval ID 298, issued on July 18, 2025**, under the project titled “*Evaluación de las Capacidades Cognitivas, de Entendimiento y Cooperación de Arquitecturas de Coordinación y Operación de Agentes Autónomos Mediados por Large Language Models*”.

The approval covers the experimental activities described in this supplementary document, including human participation in the Melting Pot environments (Commons Harvest, Externality Mushrooms, and Coins), the communication and non-communication conditions, and the resilience-disruption scenarios. All

Table S2: Baseline Extraction Summary in 1000 time-step environment

Metric	Hum.	Hum. + Comm.	Baseline
Apples <i>per capita</i>	3.40	9.20	6.30
Trees <i>per capita</i>	0.67	1.33	1.00
Collective Satiation	0.05	0.08	0.06
Equality Gini Index	0.86	0.81	0.84

participants were adults (18 years or older), provided informed consent prior to participation, and were informed that audio and screen recordings would be collected only in the communication conditions.

Moreover, the approved protocol ensures that all data are anonymized and used exclusively for academic research. Participants were free to withdraw at any time without penalty. Any modification to the originally approved activities would require submitting an updated protocol to the Ethics Committee.

References

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